FI362 — Advanced Financial Econometrics

Mock Final Examination

17 July 2026 • Duration: 2 hours • Total: 100 marks

Examination instructions — read carefully

1. **Time allowed: 2 hours.** The examination begins and ends only when the invigilator instructs.
2. **No electronic devices of any kind** may be used or kept within reach — this includes mobile phones, smart watches, tablets, laptops, headphones, **and calculators**. **Using or possessing any such device during the exam will result in immediate disqualification.**
3. **You may bring one A5 (half of A4) handwritten cheat sheet.** No printed or photocopied material, and nothing else, is permitted.
4. All numerical constants you need are provided in the **table of statistical values** at the start of Section B. Answers require only pen-and-paper arithmetic.
5. Write all **multiple-choice answers** on the Answer Sheet (page 2). Only the Answer Sheet is graded for Section A.
6. For **Question 11**, write each final answer in the box provided on the Answer Sheet *and* show your full working in the space beneath the question — working may earn partial credit.
7. Round to **two decimal places** where a value is not a whole number.
8. Do not detach any pages. Remain seated until all scripts are collected.

Marks: Section A — 10 multiple-choice questions × 7 marks = 70. Section B — Question 11 = 30 marks. **Total 100.**

Candidate details — *complete in block capitals*

Family name / Surname: ____________________

Given name: ____________________

Student ID number: ____________________

Examination room number: ____________________

Date & session (time): ____________________

Candidate signature: ____________________

Do not turn this page until you are told to begin.

ANSWER SHEET

Name: ____________________ Student ID: ____________________

Section A — Multiple Choice

Circle one letter per question. 7 marks each.

#	Choice	#	Choice
Q1	(A) (B) (C) (D)	**Q6**	(A) (B) (C) (D)
Q2	(A) (B) (C) (D)	**Q7**	(A) (B) (C) (D)
Q3	(A) (B) (C) (D)	**Q8**	(A) (B) (C) (D)
Q4	(A) (B) (C) (D)	**Q9**	(A) (B) (C) (D)
Q5	(A) (B) (C) (D)	**Q10**	(A) (B) (C) (D)

Section B — Question 11

Final answers here; full working beneath the question. 30 marks.

Part	Final answer	Marks
(a)		4
(b)		5
(c)		6
(d)		7
(e)		8

Section A — Multiple Choice

Each question is worth 7 marks. Choose the single best answer and mark it on the Answer Sheet.

Q1. The ARCH–LM test is applied to the residuals of a fitted conditional-mean model. A significant test statistic indicates that:

(A) the residuals contain a unit root;

(B) the *squared* residuals are autocorrelated — i.e. conditional heteroskedasticity (ARCH effects) is present;

(C) the residuals are normally distributed;

(D) the autoregressive order of the mean model is too low.

Q2. For a stationary ARCH(1) process with Gaussian innovations, the *unconditional* distribution of the returns:

(A) is exactly normal;

(B) has positive excess kurtosis (fatter tails than the normal);

(C) has negative excess kurtosis;

(D) has a non-zero mean equal to α_0.

Q3. A GARCH(1,1) model is estimated as $\sigma_t^2 = 0.02 + 0.08\, a_{t-1}^2 + 0.90\, \sigma_{t-1}^2$. The *persistence* of volatility is:

(A) 0.08;

(B) 0.90;

(C) 0.98;

(D) 1.00.

Q4. In a stationary GARCH(1,1), the multi-step-ahead conditional-variance forecast:

(A) stays constant at the current conditional variance;

(B) diverges to infinity;

(C) mean-reverts toward the unconditional (long-run) variance;

(D) equals zero for all horizons beyond one day.

Q5. In a two-state *Markov-switching* model, the regime (state) is:

(A) observed directly as a threshold on a lagged return;

(B) hidden (latent) and inferred probabilistically from the data;

(C) fixed and known in advance;

(D) determined by the sign of the current return.

Q6. A portfolio's 1-day 99% Value-at-Risk is \$2 million. This means:

(A) the maximum possible loss tomorrow is \$2 million;

(B) the average daily loss is \$2 million;

(C) on about 1 day in 100 the loss is expected to *exceed* \$2 million;

(D) the loss will be exactly \$2 million on 1% of days.

Q7. A Generalized Pareto Distribution is fitted to threshold exceedances of losses, and the shape parameter is estimated at $\hat{\xi} = 0.25 > 0$. This implies the loss tail is:

(A) thin, like the normal (Gumbel type);

(B) bounded above (Weibull type);

(C) heavy / power-law (Fréchet type), with tail index $1/\hat{\xi} = 4$;

(D) exactly exponential.

Q8. The extremal index of a stationary return series is estimated at $\hat{\theta} = 0.5$. This indicates that:

(A) extremes occur independently, one at a time;

(B) extremes cluster, with mean cluster size $1/\hat{\theta} = 2$;

(C) the one-day marginal VaR must be doubled;

(D) the series has no extreme values.

Q9. For a call option, the *delta* (Δ):

(A) is always greater than 1;

(B) gives the number of shares to hold to hedge the option, and lies between 0 and 1;

(C) measures sensitivity to volatility;

(D) is largest for deep out-of-the-money options.

Q10. Put–call parity states $C - P = S - Ke^{-rT}$. Its main implication is that:

(A) calls are always worth more than puts;

(B) once you can value a call, the put with the same strike and expiry is fixed by no-arbitrage;

(C) options cannot be hedged;

(D) volatility does not affect option prices.

Section B — Question 11 (30 marks)

Statistical values provided (you need no calculator):

$\Phi^{-1}(0.99)$	$= 2.326$	$\sqrt{1.64}$	≈ 1.281
$\Phi^{-1}(0.95)$	$= 1.645$	2.326×1.281	≈ 2.98
standardised t_6 1% quantile	≈ 2.57	2.57×1.281	≈ 3.29

Setup. Daily returns follow $r_t = \mu + a_t$ with $a_t = \sigma_t \varepsilon_t$, ε_t i.i.d. mean 0, variance 1, and a GARCH(1,1) conditional variance

$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

The fitted estimates (returns in %, so variances in $\%^2$) are

$$\hat{\alpha}_0 = 0.02, \qquad \hat{\alpha}_1 = 0.08, \qquad \hat{\beta}_1 = 0.90, \qquad \mu = 0.$$

Yesterday the shock was $a_{t-1} = -3.0\%$ and the conditional variance was $\sigma_{t-1}^2 = 1.00$.

(a) [4 marks] State the condition for the GARCH(1,1) to be (covariance) stationary, and compute the persistence of volatility for these estimates. Is the process stationary?

(b) [5 marks] Derive the unconditional (long-run) variance of a_t in terms of $(\alpha_0, \alpha_1, \beta_1)$, and evaluate it. State the implied long-run volatility.

(c) [6 marks] Using yesterday's information, compute today's conditional variance σ_t^2 and conditional volatility σ_t.

(d) [7 marks] (i) Compute the 1-day 99% VaR for today (as a % of the position, $\mu = 0$, conditional normality). (ii) Derive the ℓ-step-ahead variance forecast $\sigma_t^2(\ell)$ and show it converges to the unconditional variance as $\ell \to \infty$. What property does this illustrate?

(e) [8 marks] Backtesting the Normal-based 99% VaR over 1000 days produced **25** exceedances, against the 10 expected. (i) What does this indicate about the innovation distribution, and is the Normal VaR too high or too low? (ii) You re-estimate with Student-t innovations ($\nu = 6$). Using the provided standardised-t 1% quantile 2.57, compute today's 99% VaR and compare it with part (d)(i). (iii) In one or two sentences, how would a **GARCH–EVT** approach treat the tail differently, and why might it be preferred at the 99.9% level?

End of examination. Check that your Answer Sheet is complete.